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Physical Inactivity, Insufficient Sleep, and Obesity Across California Counties

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***Abstract*—This study examined how the prevalence of physical inactivity and insufficient sleep relates to adult obesity across all 58 counties in California. Age-adjusted prevalence estimates were drawn from the Centers for Disease Control and Prevention (CDC) PLACES dataset and analyzed with Pearson correlations and hierarchical multiple regression. Both physical inactivity ($r = 0.77$) and insufficient sleep ($r = 0.63$) were positively correlated with obesity ($p < 0.001$). When entered together, the two variables explained 64.0% of the variance in county obesity rates, and physical inactivity ($\beta = 0.61$, $p < 0.001$) was more strongly associated with obesity than insufficient sleep ($\beta = 0.26$, $p = 0.012$). The association for physical inactivity held across all four sensitivity analyses. The association for insufficient sleep, however, was no longer significant once smoking prevalence was controlled for or when the model was applied to all U.S. counties. These findings suggest that community-level obesity prevention should give priority to increasing leisure-time physical activity among residents.**

***Index Terms*—CDC PLACES, California, county-level analysis, obesity, physical activity, sleep.**

I. Introduction

A. Background

Obesity is a major public health problem that raises the risk of many chronic diseases, including type 2 diabetes, hypertension, and cardiovascular disease. Two lifestyle factors most often linked to obesity are physical inactivity and insufficient sleep. The U.S. Department of Health and Human Services recommends that adults get at least 150 minutes of moderate-intensity physical activity each week [1], and the American Academy of Sleep Medicine and the Sleep Research Society recommend that adults sleep at least 7 hours per night [2]. At the individual level, a meta-analysis found that people who sleep less are at higher risk of obesity [3].

Health outcomes, however, differ not only between individuals but also between places. Even within a single state, communities vary in their living environments, income levels, and access to health care, and these differences can lead to differences in residents' health. Understanding these relationships at the community level can help show which health policies are needed and where.

California is made up of 58 counties that range from large coastal metropolitan areas such as the San Francisco Bay Area to the agricultural Central Valley and the mountainous north. This diversity makes the state a good setting for examining how lifestyle factors relate to obesity at the county level.

B. Objectives and Hypotheses

The purpose of this study was to examine how the prevalence of physical inactivity and insufficient sleep relates to obesity across California counties, and to determine which of the two is more strongly associated with obesity. Four hypotheses were tested:

H1: Counties with higher rates of physical inactivity will have higher obesity rates.

H2: Counties with higher rates of insufficient sleep will have higher obesity rates.

H3: Physical inactivity and insufficient sleep will each remain significantly associated with obesity after controlling for the other.

H4: Physical inactivity will be more strongly associated with obesity than insufficient sleep.

II. Methods

A. Data

This study used the PLACES (Population Level Analysis and Community Estimates) dataset published by the CDC. PLACES provides county-level health estimates derived from the Behavioral Risk Factor Surveillance System (BRFSS) survey using multilevel regression and poststratification [4], [5]. All 58 California counties were included in the analysis, and no values were missing.

Because the age structure of the population differs from county to county, counties with a larger share of older residents can look less healthy than they actually are. To correct for this, age-adjusted prevalence was used for every variable.

B. Variables

Table I lists the variables used in the analysis, along with their definitions and data years.

TABLE I
STUDY VARIABLES

Role	Variable	Definition	Year
Dependent	Obesity (OBESITY)	Adults with a body mass index (BMI) of 30 or higher (%)	2023
Independent	Physical inactivity (LPA)	Adults who reported no leisure-time physical activity in the past month (%)	2023
Independent	Insufficient sleep (SLEEP)	Adults who sleep less than 7 hours per night on average (%)	2022
Control (sensitivity analysis)	Smoking (CSMOKING)	Adults who currently smoke cigarettes (%)	2023

C. Statistical Analysis

All analyses were performed in IBM SPSS Statistics, with the significance level set at 0.05. The analysis proceeded in four steps.

1) *Descriptive statistics:* The mean, standard deviation, minimum, and maximum were calculated for each variable.

2) *Correlation analysis:* Pearson correlation coefficients were used to examine the relationships among the three main variables (H1, H2).

3) *Hierarchical multiple regression:* With obesity as the dependent variable, physical inactivity was entered in Step 1 and insufficient sleep in Step 2. The change in R^2 was used to test how much additional variance insufficient sleep explained, and the standardized coefficients (β) of the two predictors were compared (H3, H4). Multicollinearity was checked with the variance inflation factor (VIF).

4) *Sensitivity analyses:* Four additional analyses were run to check whether the results depended on particular conditions: (a) excluding the 14 counties (top 25%) with the widest 95% confidence intervals for the obesity estimate; (b) excluding San Francisco, which was identified as an outlier; (c) adding smoking prevalence as a control variable; and (d) applying the same model to all 2,956 U.S. counties.

III. RESULTS

A. Descriptive Statistics

The mean obesity rate across California counties was 29.98%, and the county with the highest rate (37.9%) had more than twice the rate of the county with the lowest (17.1%). The mean prevalence of insufficient sleep was 35.30%, which means that more than one in three adults did not get the recommended 7 hours of sleep (Table II).

TABLE II
DESCRIPTIVE STATISTICS OF MAIN VARIABLES ($N = 58$)

Variable	Mean (%)	*SD*	Min (%)	Max (%)
Obesity	29.98	4.28	17.1	37.9
Physical inactivity	24.06	4.33	16.6	35.1
Insufficient sleep	35.30	2.41	28.7	41.5

SD = standard deviation.

B. Correlation Analysis

Physical inactivity showed a strong positive correlation with obesity ($r = 0.772$, $p < 0.001$), and insufficient sleep was also positively correlated with obesity ($r = 0.633$, $p < 0.001$). H1 and H2 were therefore supported. Physical inactivity and insufficient sleep were also correlated with each other ($r = 0.603$, $p < 0.001$), as shown in Table III. Since the two predictors were related, multiple regression was used to test whether each one was independently associated with obesity.

TABLE III
CORRELATIONS AMONG MAIN VARIABLES ($N = 58$)

Variable	1	2	3
1. Obesity	1		
2. Physical inactivity	0.772***	1	
3. Insufficient sleep	0.633***	0.603***	1

***$p < 0.001$.

Fig. 1 shows the relationship between physical inactivity and obesity. Most counties sit close to the regression line, but San Francisco falls well below it.

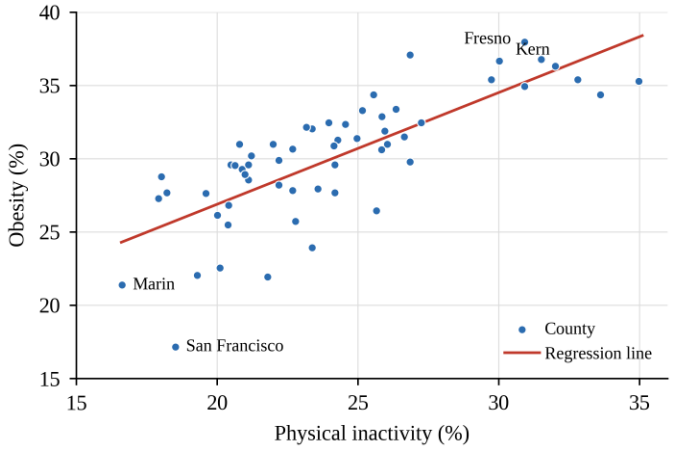


Fig. 1. Physical inactivity and obesity in 58 California counties (CDC PLACES, age-adjusted prevalence). The line is the simple regression fit (obesity = 11.64 + 0.76 × physical inactivity).

C. Hierarchical Regression

In Model 1, physical inactivity explained 59.6% of the variance in obesity, $F(1, 56) = 82.65$, $p < 0.001$. Adding insufficient sleep in Model 2 raised the explained variance to 64.0%, and this increase ($\Delta R^2 = 0.044$) was statistically significant, $F(1, 55) = 6.72$, $p = 0.012$ (Table IV).

TABLE IV
HIERARCHICAL REGRESSION PREDICTING OBESITY ($N = 58$)

	Model 1 *B* (*SE*)	β	Model 2 *B* (*SE*)	β
(Constant)	11.639*** (2.049)		−1.088 (5.282)	
Physical inactivity	0.762*** (0.084)	0.772	0.606*** (0.100)	0.614
Insufficient sleep			0.467* (0.180)	0.263
R^2	0.596		0.640	
Adjusted R^2	0.589		0.627	
F	82.65***		48.91***	
ΔR^2			0.044*	

B = unstandardized coefficient; *SE* = standard error; β = standardized coefficient. VIF for Model 2 = 1.57. *$p < 0.05$, ***$p < 0.001$.

In Model 2, physical inactivity (β = 0.614, $p < 0.001$) and insufficient sleep (β = 0.263, $p = 0.012$) were both significantly associated with obesity while controlling for each other, so H3 was supported. The standardized coefficient for physical inactivity was about 2.3 times larger than that for insufficient sleep, which supports H4. The VIF was 1.57, far below the common threshold of 10, so multicollinearity was not a concern.

The fitted equation for Model 2 is

$$\text{Obesity} = -1.088 + 0.606 \times \text{LPA} + 0.467 \times \text{SLEEP} \quad (1)$$

Put simply, among counties with the same rate of insufficient sleep, a county with a physical inactivity rate 1 percentage point higher had an obesity rate that was 0.61 points higher on average. Among counties with the same rate of physical inactivity, a 1-point higher rate of insufficient sleep was associated with a 0.47-point higher obesity rate.

D. Geographic Distribution

Most of the counties with the highest obesity rates were inland, while every county with one of the lowest rates was in the San Francisco Bay Area. Counties with high obesity also tended to have high physical inactivity (Table V).

TABLE V
COUNTIES WITH THE HIGHEST AND LOWEST OBESITY RATES

Rank	County	Obesity (%)	LPA (%)
Highest obesity			
1	Fresno	37.9	31.0
2	Riverside	37.0	26.9
3	Madera	36.7	31.6
4	Stanislaus	36.6	30.1
5	Kern	36.2	32.1
Lowest obesity			
1	San Francisco	17.1	18.5
2	Marin	21.3	16.6
3	Santa Clara	21.9	21.8
4	San Mateo	22.0	19.3
5	Alameda	22.5	20.1

LPA = physical inactivity.

San Francisco had a standardized residual of −3.14. Its observed obesity rate was about 8 percentage points lower than the value predicted by the regression equation (about 25.3%). Orange County also had a lower obesity rate than predicted, with a standardized residual of −2.30.

E. Sensitivity Analyses

Physical inactivity remained significant at $p < 0.001$ in all four sensitivity analyses. Insufficient sleep stayed significant in analyses (a) and (b), but not when smoking was controlled for (c) or when all U.S. counties were analyzed (d), as shown in Table VI.

TABLE VI
RESULTS OF SENSITIVITY ANALYSES

Analysis	*N*	R^2	LPA β	SLEEP β
Base model	58	0.640	0.614***	0.263*
(a) Excluding counties with wide CIs	44	0.662	0.591***	0.295*
(b) Excluding San Francisco	57	0.652	0.625***	0.260*
(c) Controlling for smoking	58	0.763	0.458***	0.115 (p = 0.193)
(d) All U.S. counties	2,956	0.556	0.724***	0.030 (p = 0.079)

CI = confidence interval. In (c), β for smoking = 0.444 ($p < 0.001$). *$p < 0.05$, ***$p < 0.001$.

IV. DISCUSSION

A. Interpretation of Findings

The clearest finding of this study is the link between physical inactivity and obesity. On its own, physical inactivity accounted for about 60% of the variation in obesity rates across California counties. The association remained significant when counties with wide confidence intervals or the outlier were removed, when smoking was controlled for, and when the analysis was extended to the entire United States. An association of similar size across all U.S. counties (β = 0.724) indicates that the relationship is not unique to California. This fits with existing guidelines that treat physical activity as central to preventing obesity [1].

The results for insufficient sleep should be interpreted more cautiously. Insufficient sleep was significant in the base California model, but it lost significance once smoking was controlled for, and it had almost no association with obesity across all U.S. counties (β = 0.030). This suggests that the result may not reflect a direct link between sleep and obesity. Instead, places where many people sleep too little may also be places where other health risk behaviors, such as smoking, are common.

An individual-level meta-analysis found that shorter sleep was associated with a higher risk of obesity [3]. The weaker relationship observed here may come from the difference in the unit of analysis, since a relationship seen in individuals does not always show up with the same strength in area-level averages.

The contrast between the Bay Area and inland counties was striking. Yet these regions probably differ not only in physical inactivity but also in income, education, and the urban environment, and none of these factors were measured in this study. Part of the association between physical inactivity and obesity may therefore reflect these other factors.

San Francisco's obesity rate, about 8 points lower than predicted, also deserves attention. The physical inactivity measure used here covers only leisure-time activity, so walking during commutes or other daily travel is not counted. The large amount of walking and public transit use that is typical of a dense city may help explain the gap, but this is a hypothesis that the present data cannot test.

B. Limitations

1) *Possible ecological fallacy:* This study analyzed county-

level rates, not individuals. The results cannot support individual-level claims such as "people who do not exercise are obese"; they describe relationships at the area level only [6].

2) *No causal inference:* Because the data come from a single point in time, cause and effect cannot be separated. Physical inactivity may lead to obesity, but obesity may also make exercise harder or lower sleep quality.

3) *Model-based estimates:* PLACES values are not direct surveys of each county but estimates produced by a statistical model [4]. Indicators estimated with the same method may appear more strongly correlated than they actually are.

4) *Limited control variables:* Socioeconomic factors such as income, education, racial and ethnic composition, and degree of urbanization were not controlled for.

5) *Different data years:* The insufficient sleep data are from 2022, while the obesity and physical inactivity data are from 2023.

6) *Small sample and non-normal residuals:* With only 58 counties, the number of predictors that could be included was limited. The regression residuals also did not fully follow a normal distribution (Shapiro–Wilk $p = 0.002$). Even so, the results changed very little when the outlier, San Francisco, was excluded.

V. Conclusion

An analysis of 58 California counties found that physical inactivity had the strongest and most consistent association with obesity. Insufficient sleep was also related to obesity, but that relationship weakened when other factors were controlled for or the scope of the analysis changed.

These results suggest that community-level obesity prevention policies should give priority to increasing residents' leisure-time physical activity. In inland counties where both obesity and physical inactivity are high, making activity easier through parks, recreation facilities, and community exercise programs may be especially helpful.

Future research should take socioeconomic factors such as income and education into account and use individual-level data to examine the links among physical inactivity, insufficient sleep, and obesity more directly.

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